i) Let, **w** be the subset of vector space of 2x2 matrices.

$$w=\left\{\left.\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}\right| a+b=5\right\}$$

Check whether **w** is subspace under the inherited operations.

b) If $v_1=\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $v_2=\begin{pmatrix} 1 \\ -1 \end{pmatrix}$, find w_1 and w_2.

c) Find $[v]_{B_1}$ if $[v]_{B_2}=\begin{pmatrix} 6 \\ 11 \end{pmatrix}$.

Hint: $[v]_{B_2}=[I]_{B_1}^{B_2}[v]_{B_1}$